SKYFORCE

Radar-Cued Net-Capture Interceptor UAV — Preliminary Design Report

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Organiser	P.E.G.A.S.U.S. with S.K.E.T.C.H. and CCGP IIST
Problem statement	Anti-Drone Interceptor: detect, track and safely neutralise a rogue drone near an airport or event; target recovered intact
Crew	*[Yuvraj Singh Tomar (Team Leader)]*, *[Dipanshu Kumar Kushwaha (CAD / 3D Modeller)]*, *[Shaurya Singh]*, *[Aditya Kaundilya]*
Report date	11 October 2026
Status	Preliminary design with 2-D point-mass simulation. CAD figures and CAD data are marked as placeholders and will be inserted later.

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1 Executive Summary

SKYFORCE is an electric hexarotor interceptor cued by a ground radar. It carries a CO_2-fired net (4 m opening, 5–15 m design firing window) and a 2.5 m parachute. After a geometric capture candidate it either releases the target on the spot (inside a designated safe area) or tows it to the nearest GPS safe point at low speed. The sizing mass is 18.0 kg (23.5 kg while towing a 5 kg target), under the 25 kg limit.

Table 1: Key design figures (preliminary; Section 5 shows the derivations and which values are assumed).

Parameter	Value	Parameter	Value
Configuration	Hexarotor, 28 in propellers	Sizing / tow mass	18.0 kg / 23.5 kg
Motor / ESC	T-Motor U8 Lite KV100 / Alpha 60A 12S	Max static thrust, T/W	429.7 N, 2.43
Hover / sprint power	1.56 kW / 3.61 kW (25 m/s)	Energy ceiling	207 Wh (no tow), 253 Wh (60 s tow)
Battery	12S 8 Ah, 355 Wh nominal, 284 Wh usable	Arm / span	0.800 m / 2.31 m
Arm bending safety factor	10.9 (simplified)	Canopy / descent rate	2.5 m / 4.74 m/s
Launch response threshold	33.8 s	Cost (1 drone, no spares)	INR 598,037

Simulation outcome. Eight scenarios were run in a MATLAB 2-D point-mass model: three post-launch mission cases (attack, defense, tow), two gate-only no-takeoff cases, a proportional-navigation (PN) diagnostic, an overweight-target policy case and an extended-pursuit case. Geometric capture candidates were reached in 5–15 m windows in all five flown candidate cases. In every flown case the speed stayed within 25 m/s and the applied acceleration within 4.5 m/s^2. The extended-pursuit case is stopped by the energy guard (projected 307.8 Wh against a 253 Wh ceiling), which shows that reaching the window is not the same as completing the mission.

Scope statement. All results are model-state outcomes based on assumed inputs. No physical net capture, parachute deployment, forward-flight thrust or battery discharge is demonstrated or claimed. A 5–15 m event is called a *geometric capture candidate* throughout.

2 Problem Understanding and Assumptions

2.1 Mission and requirements

The brief is to detect, track and safely neutralise a rogue drone near an airport or event, with the target recovered intact. The design must justify the capture mechanism, the guidance approach and how collateral risk is kept minimal. The requirements the design and simulation respect are:

- net capture with parachute-assisted recovery of targets up to 5 kg;
- sprint speed of at least 25 m/s;
- detection range of at least 1 km using a stated sensor (ground radar);
- vision or radar terminal guidance;
- takeoff weight under 25 kg.

Deliverables are the design report with sizing calculations, CAD, analysis/simulation results, a budget and a cost estimate.

2.2 Principal assumptions

- Sea-level ISA ($\rho = 1.225$ kg/m^3, $g = 9.81$ m/s^2); figure of merit $FM = 0.65$; drag $C_D = 1.2$, frontal area 0.15 m^2 (all assumed).
- Reaction delay of 1.8 s, stacked from radar update (≈0.5 s), GCS planning (0.3–0.5 s), link latency (≤0.2 s) and spin-up/launch decision (0.5–1 s). This is a working assumption, not a measured figure.
- Simulation caps of 25 m/s speed and 4.5 m/s^2 applied acceleration; a deliberate control-smoothness limit, far below the 21.8 m/s^2 static-available acceleration.
- Two boundaries: outer tracking ring of 2,000 m and inner protected circle of 1,000 m radius. Safe points use a 120 m radius. These are illustrative scenario settings, not approved operational limits.
- Sizing mission: 120 s hover-equivalent plus 120 s sprint at 25 m/s, 20% demand margin, and a 60 s tow phase for the recovery branch.

3 Avionics and Communications Architecture

3.1 Signal and control path

The radar never talks to the drone directly. It feeds the GCS laptop over a wired link. The GCS converts the track into target position and velocity in the drone frame, plans the intercept, and sends cues and commands over *one* bidirectional long-range radio link (MAVLink, 1 W class). Telemetry returns on the same link. The drone flies to the cued point, then the camera takes over for terminal guidance.

3.2 Interfaces and link budget

Free-space link budget (assumed: 30 dBm transmit power, 2 dBi antennas, −110 dBm sensitivity, about 866 MHz):

$$\mathrm{FSPL}(d) = 20\log_{10} d_{\mathrm{km}} + 20\log_{10} f_{\mathrm{MHz}} + 32.44,$$
$$1\text{ km: } 91\,\mathrm{dB} \Rightarrow \text{margin} = 30 + 2 + 2 - 91 + 110 = 53\,\mathrm{dB}; \qquad 4\text{ km: } 103\,\mathrm{dB} \Rightarrow 41\,\mathrm{dB}.$$

At 915 MHz the path loss rises by only 0.5 dB. The legal band and power for India must be confirmed before a radio model is chosen (licence-free 865–867 MHz, 2.4 and 5.8 GHz), and all assumed values must be replaced by datasheet values and a range test.

Camera detectability (1920 px over 90° field, 0.8 m target span, 21.3 px/deg): about 10 px at 100 m, 33 px at 30 m and 65 px at 15 m. The radar cue must therefore bring the drone close before the camera can lock, or a

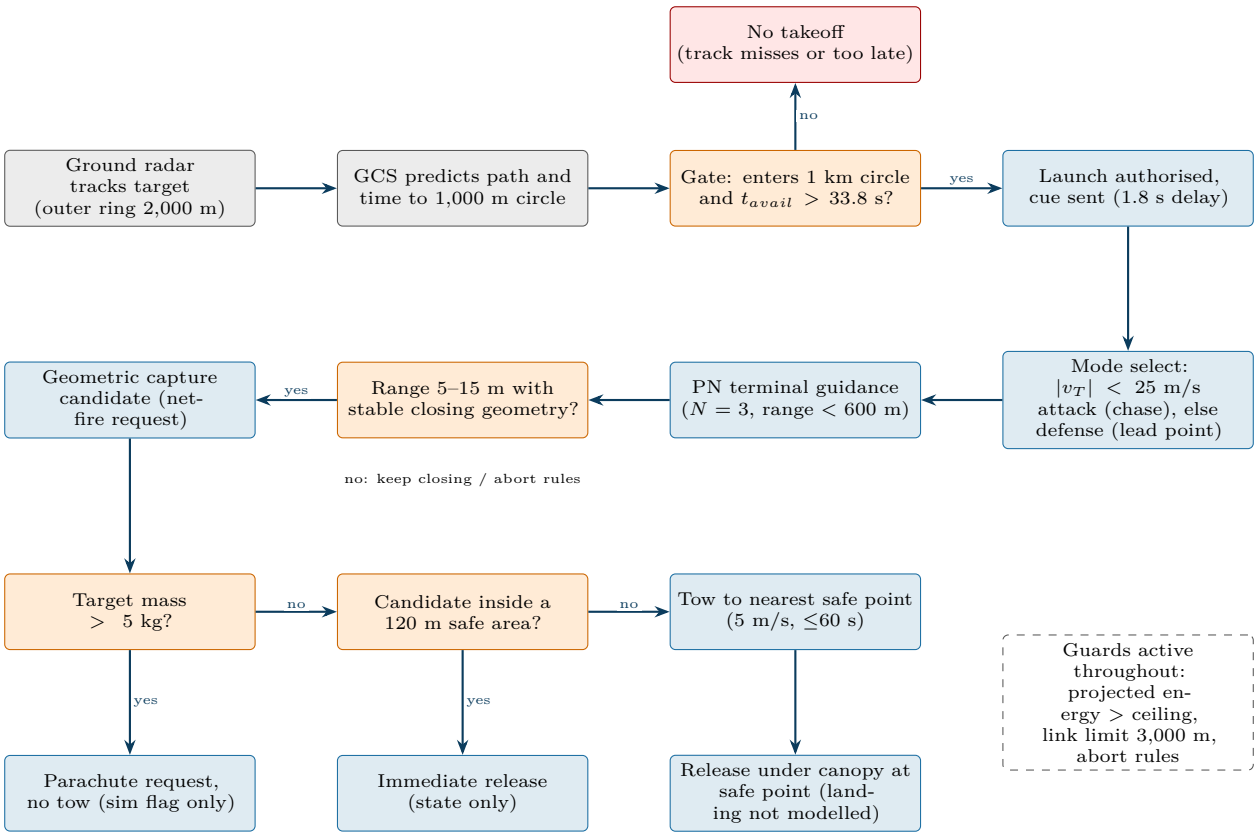


Figure 1: Concept of operations and decision logic (orange = decision, red = stop). Release, tow and parachute branches are state logic only.

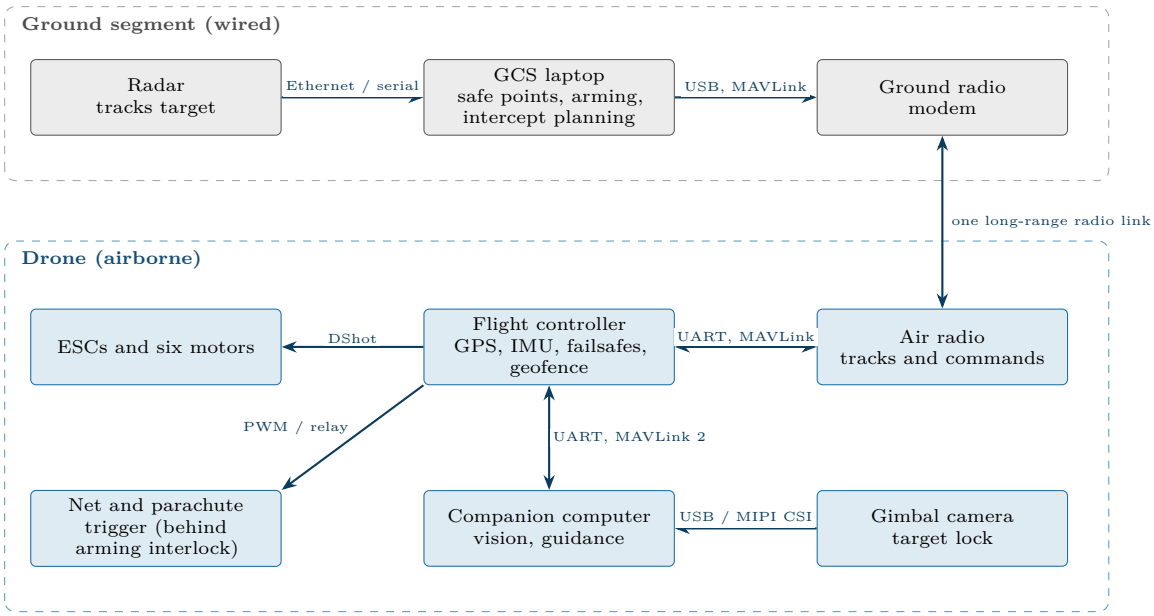


Figure 2: Block diagram 1: signal and control path. There is no second ground radio and no RC receiver in the current design.

narrower lens is needed.

Firing rules (proposed). The operator gives an explicit arm command from the GCS; flight-controller state is acceptable (geofence, battery, link); range and lock confidence are inside the 5–15 m design window; a hardware arming switch sits in the trigger circuit; and fire permission expires if the link heartbeat is lost. The companion computer runs offline from local code and model files.

4 Sizing Calculations

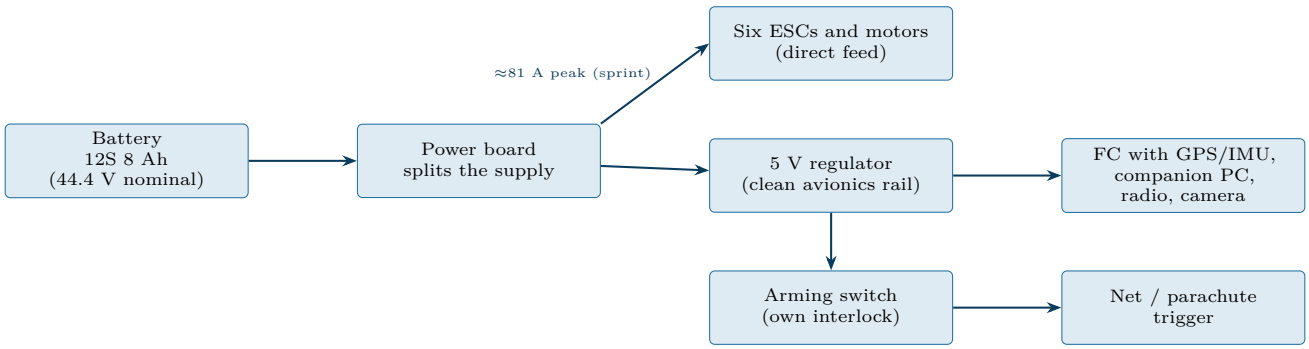


Figure 3: Block diagram 2: power distribution. Avionics sit on a separate regulated rail so motor noise cannot reset them; the trigger circuit cannot fire from a software fault alone.

4.1 Mass ledger

The 3.22 kg allowance is a named, visible allowance and is not yet closed against real parts. Several ledger lines are placeholders; bare reference masses are lower (Table 5). These are not always like-for-like assemblies, so they must be reconciled against the final BOM before any hardware claim.

4.2 Thrust and propulsion selection

$$W = mg = 18.0 \times 9.81 = 176.6\,\text{N}, \qquad T_{\max} = 6 \times 7.3 \times 9.81 = 429.7\,\text{N},$$
$$T/W = 429.7/176.6 = 2.43, \qquad T_{\text{hover}}/\text{motor} = W/6 = 29.4\,\text{N (3.0 kgf, 41\% of max thrust)}.$$

Hover at 41% of maximum thrust is not a throttle figure; throttle-to-thrust behaviour comes from bench data. The selection is T-Motor U8 Lite KV100 (12S, 27–28 in propeller), G28x9.2 CF propeller and Alpha 60A 12S ESC, with 7.3 kgf maximum static thrust per motor (published). Static bench data (reference only): 40% throttle gives 2.243 kgf, 3.5 A, 168 W; 50% gives 3.231 kgf, 6.0 A, 288 W.

4.3 Hover power (two independent estimates)

$$A = \pi R^2 = \pi(0.3556)^2 = 0.3973\,\text{m}^2, \qquad v_i = \sqrt{\frac{T}{2\rho A}} = \sqrt{\frac{29.43}{2(1.225)(0.3973)}} = 5.50\,\text{m/s},$$
$$P_{\text{ideal}} = Tv_i = 161.8\,\text{W}, \qquad P = \frac{P_{\text{ideal}}}{FM} = \frac{161.8}{0.65} = 249\,\text{W} \;\Rightarrow\; 6P = 1.49\,\text{kW}.$$

Bench interpolation: 3.0 kgf lies 76.6% of the way from 2.243 to 3.231 kgf, so $P = 168 + 0.766(288 - 168) = 260$ W per motor and $6P = 1.56$ kW. The two estimates agree within about 5%; the bench-based 1.56 kW is used (momentum theory per Leishman [**?**]).

4.4 Sprint power and mission energy

$$F_D = \tfrac{1}{2}\rho C_D A v^2 = 0.5(1.225)(1.2)(0.15)(25)^2 = 68.9\,\text{N}, \qquad \theta = \arctan(68.9/176.6) = 21.3^\circ,$$
$$P_{\text{par}} = F_D v = 1.72\,\text{kW}, \qquad P_{\text{sprint}} = (1.56 + 1.72) \times 1.10 = 3.61\,\text{kW},$$
$$T_{\text{req}} = \sqrt{W^2 + F_D^2} = 189.5\,\text{N} = 19.3\,\text{kgf} \qquad \Rightarrow 3.2\text{ kgf per motor against 7.3 kgf available.}$$

$$E_{\text{sprint}} = 3.61 \times 120/3600 = 120.3\,\text{Wh}, \qquad E_{\text{hover}} = 1.56 \times 120/3600 = 52.0\,\text{Wh},$$
$$E_{\text{mission}} = 172.3\,\text{Wh} \xrightarrow{\times 1.20} 206.8 \approx \mathbf{207}\,\text{Wh}.$$

The 3.61 kW figure adds hover power to parasite power and is therefore conservative. It rests on assumed drag values and does not prove that the airframe sustains 25 m/s.

4.5 Tow phase and battery sizing

$$m_{\text{tow}} = 18.0 + 5.5 = 23.5\,\text{kg (3.92 kgf per motor, 54\% of max)}, \qquad P_{\text{tow}} = 1.56\left(\tfrac{23.5}{18.0}\right)^{1.5} = 2.33\,\text{kW},$$
$$E_{\text{tow}} = 2.33 \times 60/3600 = 38.8\,\text{Wh} \xrightarrow{\times 1.20} 46.5\,\text{Wh}, \qquad E_{\text{total}} = 206.8 + 46.5 = \mathbf{253}\,\text{Wh}.$$

The 20% margin is applied separately to the base mission and to the tow addition, and the 80% usable-capacity convention is a second, intentionally stacked margin.

Pack current at sprint is $3610/44.4 = 81$ A (10.2C on 8 Ah), or 13.6 A per motor against a 60 A ESC rating; tow-hover pack current is 52 A. Ageing, temperature and voltage sag reduce usable energy, and the actual pack discharge rating still needs checking.

4.6 Arm and frame

$$L \geq 1.1 \times 711\,\text{mm} = 782\,\text{mm} \rightarrow 800\,\text{mm}, \qquad \text{Span} = 2(800 + 355.6) = 2311\,\text{mm},$$
$$I = \tfrac{\pi}{64}(0.035^4 - 0.031^4) = 2.83 \times 10^{-8}\,\text{m}^4, \qquad F = 7.3(9.81) + 0.238(9.81) = 73.9\,\text{N},$$
$$M = FL = 59.1\,\text{N m}, \qquad \sigma = \tfrac{Mc}{I} = \tfrac{59.1(0.0175)}{2.83\times10^{-8}} = 36.5\,\text{MPa},$$
$$\text{SF} = 400/36.5 = 10.9, \qquad \delta = \tfrac{FL^3}{3EI} = \tfrac{73.9(0.512)}{3(70\times10^9)(2.83\times10^{-8})} = 6.4\,\text{mm}.$$

The arm is a 35 mm OD, 2 mm wall carbon tube with an assumed 400 MPa allowable and $E = 70$ GPa. This is a simplified cantilever check; laminate properties, arm-root joints, fasteners, fatigue and impact are not covered, so 10.9 is not a whole-airframe safety factor. In a regular hexagon the gap between adjacent propeller discs is $800 - 711 = 89$ mm (12.5% of the diameter).

4.7 Parachute and net

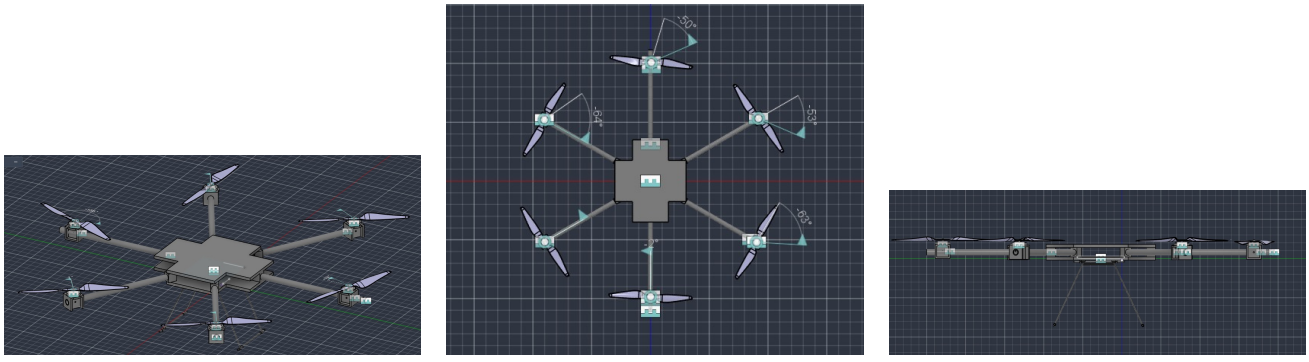
$$W_s = (5.0 + 0.5)(9.81) = 53.96\,\text{N}, \qquad A = \tfrac{2W_s}{\rho C_{Dp} v^2} = \tfrac{2(53.96)}{1.225(0.8)(5)^2} = 4.40\,\text{m}^2 \;\Rightarrow D = 2.37\,\text{m} \rightarrow 2.5\,\text{m},$$
$$v_d = \sqrt{\tfrac{2W_s}{\rho C_{Dp} A}} = 4.74\,\text{m/s}\ (A = 4.91\,\text{m}^2), \quad KE = \tfrac{1}{2}(5.5)(4.74)^2 = 62\,\text{J}.$$

$C_{Dp} = 0.8$ is assumed (parachute sizing after Knacke [**?**]). The 4.74 m/s figure is a descent rate and not a safety proof. The net opening is $3 \times 1.2 = 3.6 \rightarrow 4$ m for an assumed 1.2 m target span. Opening shock, pendulum motion, deployment reliability, net coverage, targeting error and capture probability are not evaluated.

4.8 Final dimensions and requirement check

5 CAD Model

The repository already holds the step files for individual CAD models and complete project.



(a) Isometric view **(b)** Top view **(c)** Front view

Figure 4: Three-view CAD drawing with key dimensions.

6 Software and Algorithm Design

Mode selection and long-range guidance. If the target speed is below 25 m/s the interceptor flies in attack mode; otherwise it flies in defense mode. In attack mode the aim point is the smallest positive root t^* of

$$(|v_T|^2 - v_{\max}^2)\, t^2 + 2(r \cdot v_T)\, t + |r|^2 = 0, \qquad r = p_T - p_I, \qquad p_{\text{aim}} = p_T + v_T t^*.$$

The desired speed is $v_{\text{des}} = \min\!\left(v_{\max}, \sqrt{2a_{\max} d}\right)$ along the aim direction and the reference command is $a_{\text{ref}} = 2(v_{\text{des}} - v_I)$. In defense mode the drone moves to a lead point ahead of the predicted track using a PD waypoint law $a = K_p e - K_d v$ ($K_p = 0.35\ \text{s}^{-2}$, $K_d = 1.5\ \text{s}^{-1}$, illustrative) and holds there for the target to enter the window.

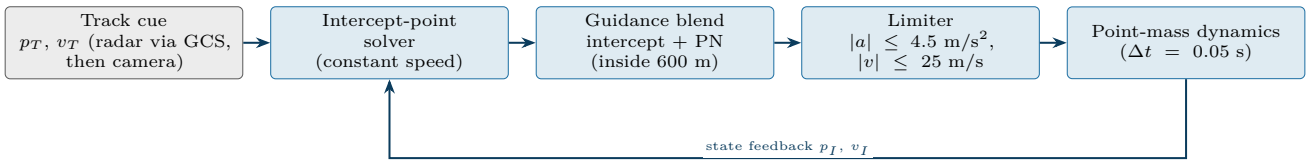


Figure 5: Block diagram 3: guidance and control loop used in the simulation.

Terminal guidance. Inside 600 m, proportional navigation (Zarchan [**?**]) is blended in:

$$a_{PN} = N\, V_c\, \dot{\lambda}\, \hat{n}, \quad V_c = -\hat{r} \cdot (v_T - v_I), \quad \dot{\lambda} = \frac{r_x v_{rel,y} - r_y v_{rel,x}}{|r|^2}, \quad a_{\text{cmd}} = 0.25\, a_{\text{ref}} + 0.75\, a_{PN},$$

with $N = 3$ and $\hat{n} \perp v_I$. Every command passes through the same limiter in all phases: $a \leftarrow \text{sat}(a_{\text{cmd}}, 4.5)$, $v \leftarrow \text{sat}(v + a\Delta t, 25)$. The closing speed reported at the candidate is the line-of-sight value $V_{close} = -\hat{r} \cdot (v_T - v_I)$, not the magnitude of the relative velocity.

Capture candidate, release and tow. A candidate is registered after the 1.8 s reaction delay when $5 \leq R \leq 15$ m (in defense mode only once the drone has settled at the lead point: within 8 m and below 1 m/s). The decision is then:

- target $\leq$5 kg and inside a 120 m safe-point radius: immediate-release state;
- target $\leq$5 kg and outside all safe areas: tow to the nearest safe point at a 5 m/s setpoint, ramp-limited, completed when the safe-area radius is entered within 60 s;
- target >5 kg: simulation-only parachute-request flag and towing disabled (overrides the tow choice).

An earlier version showed a 222.89 m/s^2 acceleration spike at the capture-to-tow transition (an unramped speed-target change). The fix synchronises the mass update at capture, starts the tow velocity target from the actual capture-instant velocity and ramps to the 5 m/s setpoint under the same 4.5 m/s^2 limiter. Hand check for 25 to 5 m/s: $\Delta V = 20$ m/s, ramp time $20/4.5 = 4.44$ s, ramp distance 66.7 m.

Two-boundary launch gate. A constant-velocity track prediction is checked against the 1,000 m inner circle:

$$t_{\text{resp}} = t_{\text{react}} + t_{\text{int}} + t_{\text{margin}} = 1.8 + 30 + 2 = 33.8\,\text{s}.$$

If the predicted track misses the circle there is no takeoff. If it enters with $t_{\text{avail}} > 33.8$ s a simulated launch is recommended; if $t_{\text{avail}} \leq 33.8$ s there is no takeoff (too late). 33.8 s is an illustrative demonstration value, not a certified launch rule. The abort rule also triggers if the target crosses the hard boundary or if time to the boundary is at most the best-case intercept time plus 2 s.

Energy and link guards. Power is estimated from a hover-power baseline scaled with mass$^{1.5}$, a sprint endpoint scaled linearly with mass, and a speed-cubed blend between them. This is an illustrative approximation and not a measured discharge model. The energy guard stops the pursuit when

$$E_{\text{projected}} = E_{\text{used}} + E_{\text{remaining}} + E_{\text{buffer}} > E_{\text{ceiling}},$$

with assumed buffers of 9.7 Wh (no tow) and 18.8 Wh (tow) for launch, descent and avionics. The communications guard enforces a 3,000 m effective link limit using both current and predicted next-step distance; all four unit-test cases pass.

7 Simulation Methods and Results

7.1 Method

The model is a MATLAB (R2018b or newer) 2-D point-mass scenario simulation with $\Delta t = 0.05$ s and a 240 s horizon. It is not flight-controller code. It exports a scenario summary, a sample-by-sample trajectory log (positions, velocities, line of sight, range rate, closing speed, requested and applied acceleration, active limiter, phase, power, cumulative energy), a launch-gate table, a propulsion-sensitivity table, a working-range table, an overweight-policy table, the mass ledger, a component-mass comparison and an assumptions list. The interceptor station is fixed on the 1,000 m reference circle while it awaits the launch decision.

Common checks for all flown cases: maximum speed 25.0 m/s (cap respected) and maximum applied acceleration 4.50 m/s^2 (4.05 m/s^2 in the PN diagnostic). Requested acceleration before limiting peaked at 50.6, 45.0 and 35.1 m/s^2 in scenarios 1, 2 and 3 (and 5.0 m/s^2 in 7 and 8), which is why the limiter matters. The maximum telemetry distance was 1.0–1.6 km, well inside the 3,000 m link limit, so the link guard never triggered. Projected energy equals simulated energy plus the estimated remaining segment plus the buffer; in scenario 3 it excludes landing and return.

7.2 Launch gate (scenarios 4 and 6)

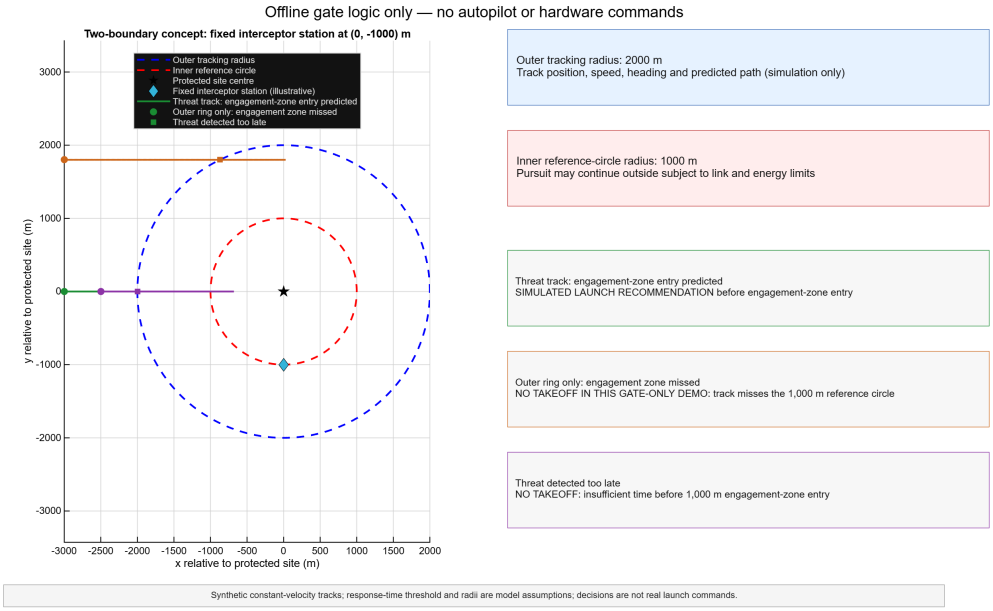


Figure 6: Two-boundary launch-gate demonstration (gate-only, no flight).

Three gate outcomes are shown. A 20 m/s track from 3,000 m gives 50 s from outer-ring acquisition to inner entry, which exceeds 33.8 s, so launch is recommended. A track that misses the 1,000 m circle (closest approach 1,800 m) gives no takeoff. A 40 m/s track from 2,500 m gives only 25 s to inner entry, so there is no takeoff. The flown abort case (scenario 4, 15.0 s available) and the outer-ring case (scenario 6) follow the same logic; the corresponding figure files `skyforce_figure_04` and `skyforce_figure_05` are in the repository.

7.3 Attack, defense and tow cases (scenarios 1–3)

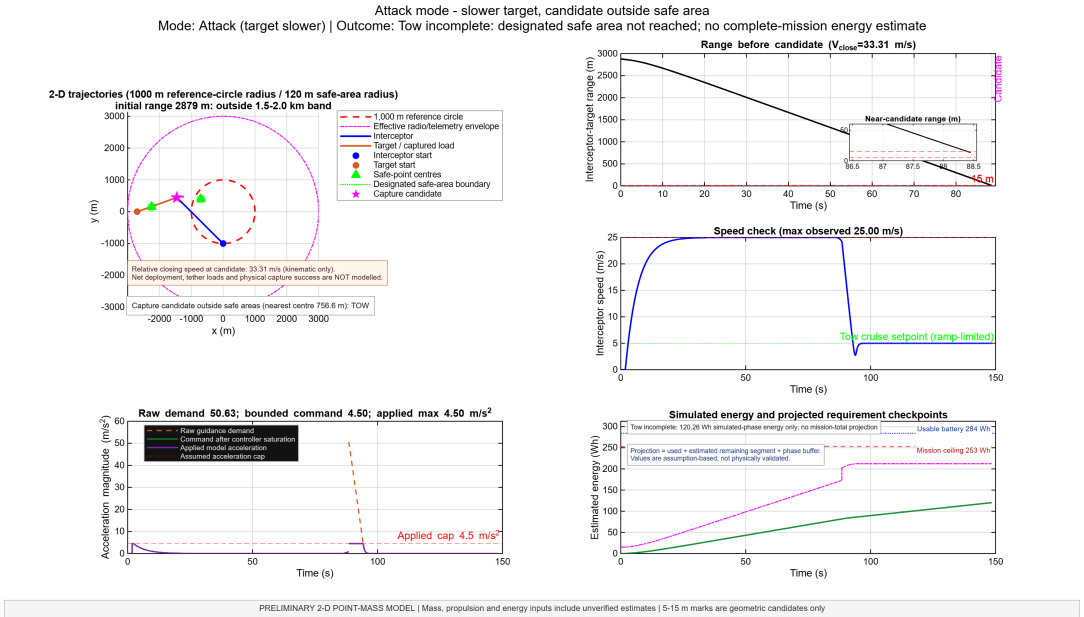


Figure 7: Scenario 1: attack mode; candidate outside safe area, tow incomplete at 60 s.

In scenario 1 the 14.87 m/s target is overtaken in tail-chase geometry and a candidate occurs at 13.35 m after 88.45 s. It lies 756.6 m from the nearest safe-point centre, so a tow is required; the tow does not reach the safe area within 60 s and the case is reported as incomplete, with no complete-mission energy claim. Scenario 2 uses a 27 m/s target, above the 25 m/s cap, so the interceptor holds at a lead point and the candidate occurs at 14.60 m, 6.9 m from a safe-point centre, giving an immediate-release state with 79.58 Wh projected against 207 Wh (38% of the ceiling). In scenario 3 the candidate (13.97 m, 82.85 s) lies 225.4 m from the nearest safe point; the tow

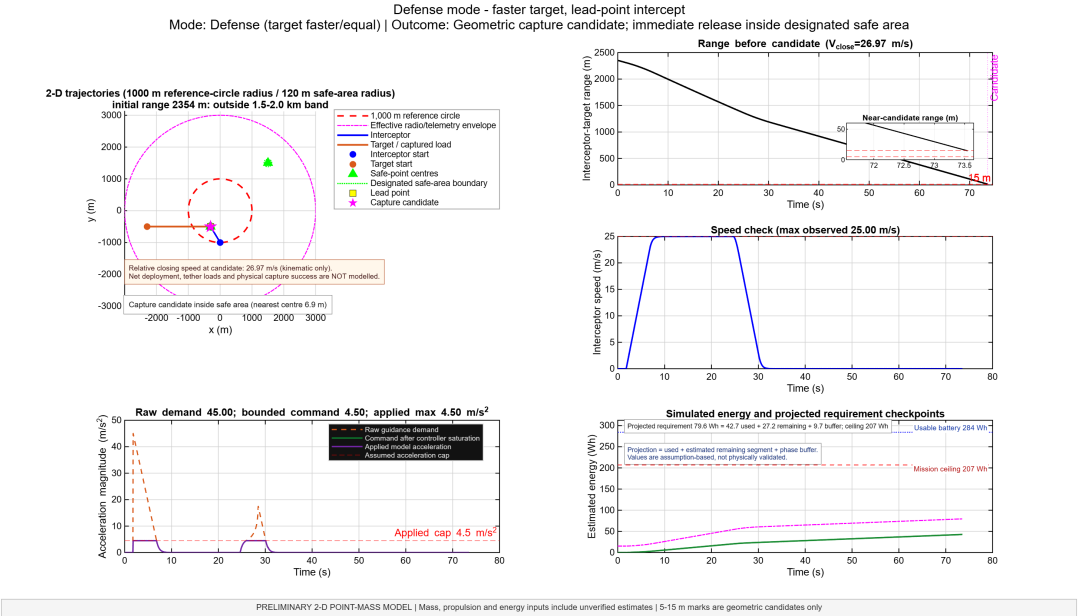


Figure 8: Scenario 2: defense mode with a 27 m/s target and lead-point intercept.

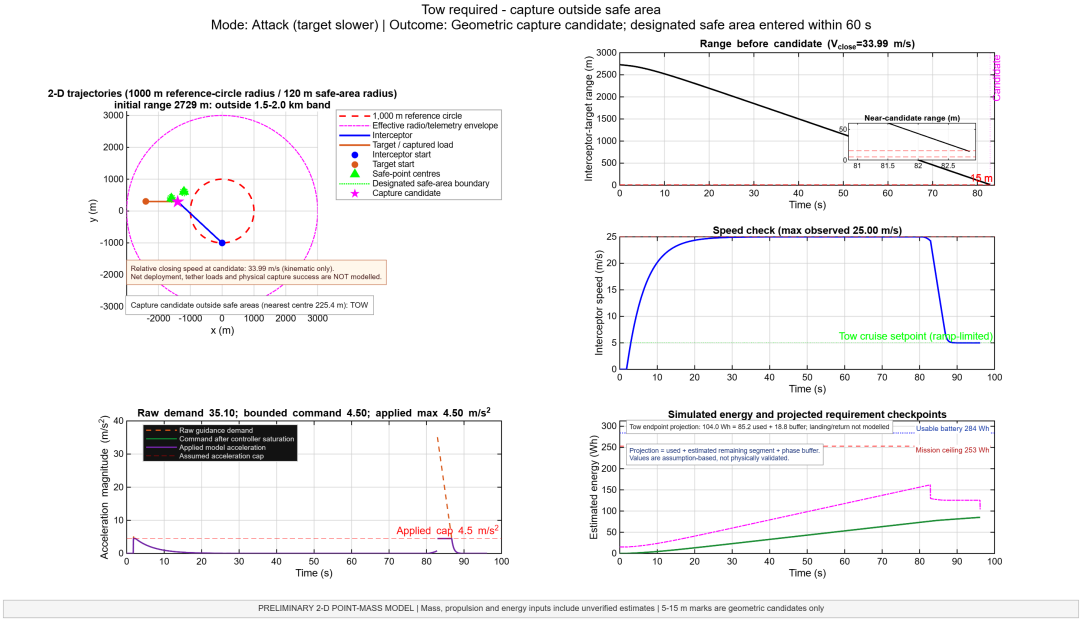


Figure 9: Scenario 3: tow required; safe area entered within the 60 s criterion.

ramps from about 25 m/s to the 5 m/s setpoint and enters the safe area after 13.3 s, giving 103.98 Wh against 253 Wh (41%), excluding landing and return.

7.4 PN diagnostic (scenario 5)

The earlier scenarios start on near-collision courses and cannot show PN correcting anything. Scenario 5 starts the interceptor at 20 m/s with a $-26.6°$ heading offset against a crossing target. PN is active for 35.4 s with a peak PN acceleration of 0.42 m/s^2, and a candidate occurs at 14.28 m with 16.09 m/s closing speed. This is a guidance diagnostic and not a mission-energy case.

7.5 Overweight target and extended pursuit (scenarios 7 and 8)

A mass above 5 kg does not cause a mass-only abort. In scenario 7 (6 kg, 1 kg over the limit) the candidate geometry matches scenario 3, after which the model sets the parachute-request flag (post-capture mass 24.0 kg) and disables towing. The 2.5 m canopy is sized for a 5.5 kg suspended load, so deployment, opening shock and recovery of a 6 kg load are not validated. In scenario 8 the gate predicts inner-zone entry with 49.97 s available against 33.8 s required, so launch is authorised. The target enters and then exits the 1 km zone at 127.26 s near

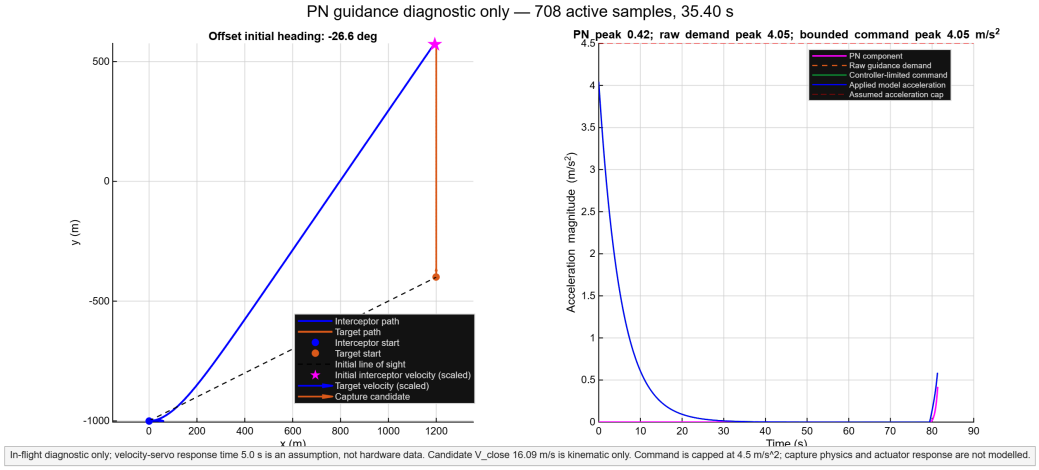


Figure 10: Scenario 5: PN correction diagnostic with a deliberate initial heading offset.

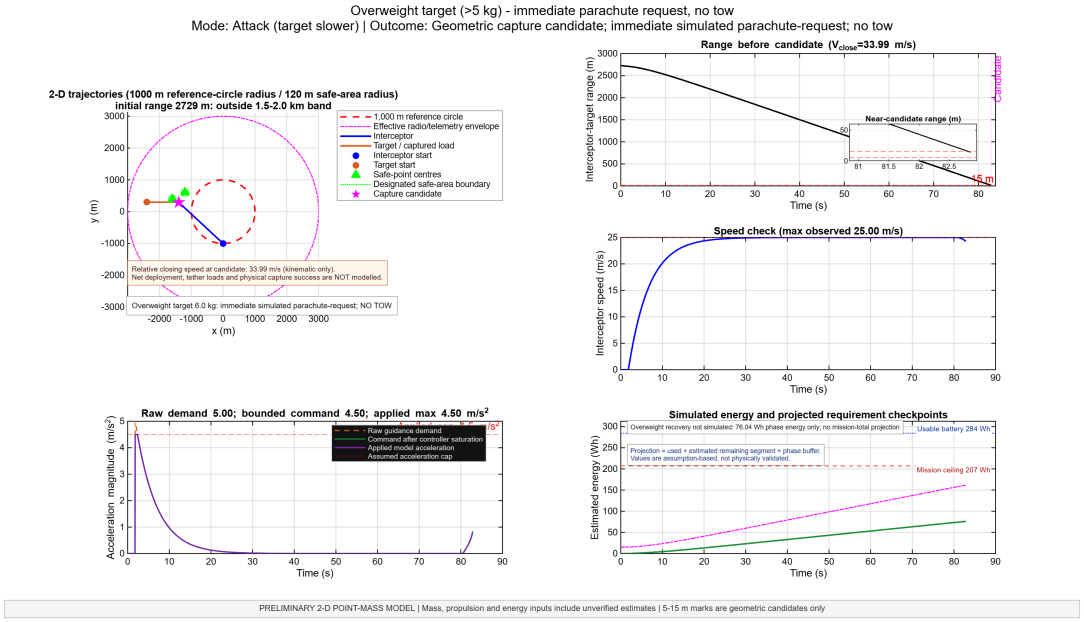


Figure 11: Scenario 7: 6 kg target; simulated parachute request and no tow.

(435.9 m, 900 m), and the 13.95 m window is reached geometrically but not registered. At the stop the projected requirement is $78.44 + 210.60 + 18.8 = 307.84$ Wh, which exceeds the 253 Wh ceiling by 54.84 Wh and the 284.16 Wh usable energy by 23.68 Wh.

7.6 Propulsion margin sensitivity

Required thrust at 25 m/s is 189.5 N (nominal) and 235.9 N (post-capture). All margins exceed 1.0, but no measured thrust-versus-airspeed curve exists for the selected propeller, so these are arithmetic checks only. The 25 m/s plotted maximum shows the model stayed under its own cap and is not proof of flight capability.

7.7 Limitations and data notes

- A 5–15 m event is a geometric capture candidate only. Net deployment, tether loads, capture probability, parachute dynamics, wind, sensor error, forward-flight propeller performance and detailed electrical losses are not modelled.
- Complete-mission energy values are estimates (simulated phases plus assumed buffers); no single continuous launch-to-landing integration exists yet.
- The simulation uses a 23.0 kg post-capture mass (18 kg plus a 5 kg target); the tow-energy sizing uses a more conservative 23.5 kg (including 0.5 kg net and tether), a difference of about 3% in tow power.
- Most flown scenarios start outside the nominal 1.5–2.0 km working-range band (2.35–3.55 km); they are treated as diagnostic exceptions pending confirmation. Only the gate-only abort case (1,640 m) lies inside the band.

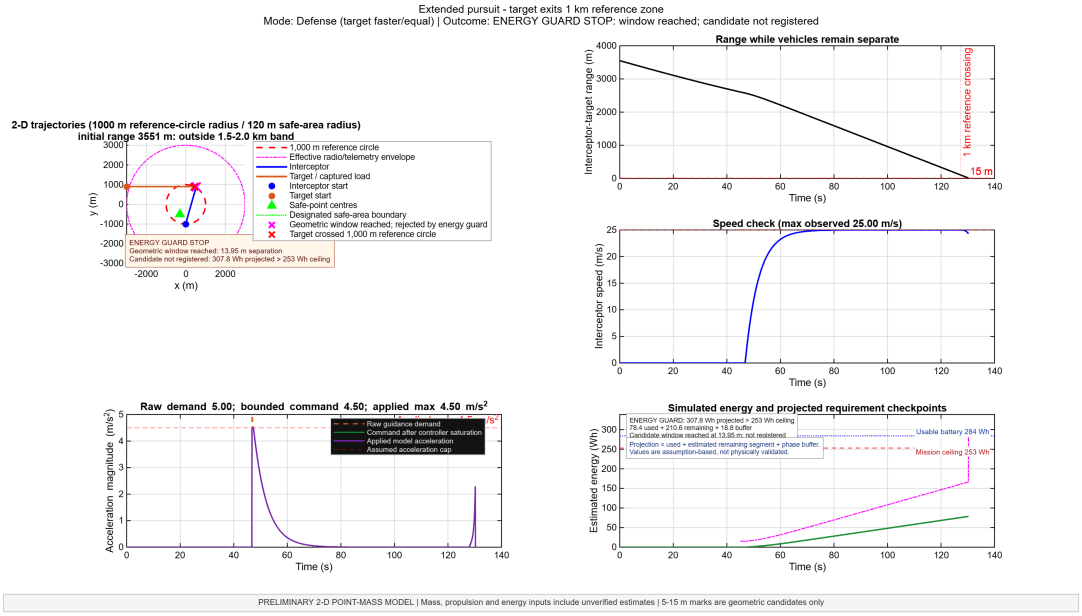


Figure 12: Scenario 8: extended pursuit stopped by the energy guard before candidate registration.

- Scenario 1 energy: the summary and trajectory exports supplied for this report agree (120.26 Wh). Both should be regenerated from one MATLAB run before final freeze.

8 Bill of Materials and Cost

By category: propulsion 361,710 (60.5%), avionics 172,727, power 30,500, capture payload 15,800, structure 14,300 and miscellaneous 3,000. Sourced listings cover INR 450,437 (75%) and estimates INR 147,600 (25%). The BOM excludes the ground segment (radar, GCS laptop, ground radio, software), shipping, customs and import duty on items not marked GST-inclusive, and spares. Two substitutions need reconciling with the mass ledger: the U8II motor is 272 g against 238 g for the U8 Lite (about 0.2 kg more over six motors), and the Flame 80A ESC replaces the specified Alpha 60A. Estimated lines need real quotes before purchase.

9 Feasibility, Scalability and Risks

Feasibility. On paper the design meets every stated requirement with margin: 3.2 of 7.3 kgf per motor at sprint, 2.43 static T/W, a 12% energy margin with the tow, and a 10.9 arm safety factor. These are sizing-level results only; capture and recovery performance remain design targets.

Scalability. The payload bay (net launcher, CO_2, net, parachute) is a modular unit, and the boundary, track-prediction and launch-gate logic is independent of the interceptor type. It could transfer to a fixed-wing or hybrid VTOL interceptor, but the point-mass dynamics could not (no hover, hold or low-speed tow). A radar and GCS can cue several pads, and site radii and safe points are software settings rather than airframe geometry. The air vehicle costs about INR 5.98 lakh, excluding the ground segment. The airframe uses standard carbon tube and a scratch-built plate frame, which keeps manufacturing simple.

10 Safety, Regulatory and Ethical Considerations

- **Safety by design.** A hardware arming switch and explicit GCS arm command gate the trigger; fire permission expires on loss of heartbeat; the flight controller keeps geofence, battery and link failsafes. Safe points are pre-loaded as GPS coordinates, and a netted target is towed to the nearest one before release rather than dropped over people.
- **Safe abort.** The design does not promise a 100% capture rate; abort rules (hard boundary and time to boundary), the launch gate, the energy guard and the 3,000 m link guard all fail towards “no engagement”. A heavier target above 5 kg triggers the parachute-request logic and no tow.
- **Regulatory.** Airspace and regulatory limits for UAV operations in India (the Drone Rules in force), the radio band, and any authorisation for counter-UAS use must be confirmed before any real deployment. The design is for lawful use by authorised operators.

- **Ethics and privacy.** Capture is non-destructive and the target is meant to be recovered intact. Camera imagery is used on board for target lock; retention and sharing would need a policy. No part of the design is intended to injure people or damage property.
- **Honest claims.** All performance figures come from our own calculations or simulation, or are cited; none are flight-tested. Assumed and withdrawn values are labelled, and superseded baselines (22 in propeller, 236 Wh, 300 m keep-out) are not reused.

11 Conclusion and Declarations

SKYFORCE meets the stated requirements at sizing level with an 18.0 kg hexarotor, a 207/253 Wh energy budget on a 355 Wh pack and a drone cost of INR 598,037. The simulation demonstrates scenario-level decision logic: launch gating, attack and defense geometry, PN correction, release-or-tow decisions, the overweight-target policy and an energy-guard stop. It does not demonstrate physical capture, recovery or measured propulsion performance. The next steps are to close the mass ledger against real parts, select a battery, replace assumed drag and thrust-loss values with calculator or test data, run one continuous launch-to-landing energy integration, and insert the final CAD.

Pre-built modules, third-party material and AI tools used are listed in the signed Annexure C declaration submitted with this package *[team to confirm]*.

References

[1] T-Motor: U8 Lite KV100 motor, G28x9.2 CF propeller and Alpha 60A 12S ESC product pages and test data (manufacturer, static reference data).

[2] Cube Orange+ autopilot, RFD900x radio and Skydroid C12 gimbal camera product information; MAVLink protocol documentation (mavlink.io).

[3] Fortem Technologies F700 and TrueView R30 radar public product information, as cited in the team's architecture notes.

[4] Retailer listings (Robu.in, xboom.in, Zbotic.in) for prices in Table of Section 8, accessed 10 October 2026.

[5] ORION UAV Design Challenge, Official Rulebook v1.0, P.E.G.A.S.U.S., 2026.

[6] Team 08 internal documents: Sizing Calculations, Communications and Wiring Architecture, Consolidated Design and Simulation Reference, Guide Decision Record, Change Log, MATLAB source and CSV/figure exports (10–11 October 2026).

Table 2: Main design decisions and the reasons for them.

Decision	Chosen	Reason / alternative not chosen
Airframe	Electric hexarotor	Must hover at a lead point, slow to a 5 m/s tow speed and hold waypoints. A fixed-wing aircraft cannot do these and would need stall, turn-radius and bank-angle models; a hybrid VTOL needs separate hover, transition and cruise dynamics. Six 7.3 kgf motors give T/W 2.43 (1.86 with the target). Motor-out tolerance is not assessed.
Propulsion	U8 Lite KV100 with 28 in G28x9.2 CF	Published thrust data exists for this pairing. An earlier 22 in baseline had no matching published data and was withdrawn. The span grows from 1.86 m to 2.31 m (reference: 1.98 m for the Fortem F700).
Sensing	Ground radar cue, then onboard gimbal camera	Radar gives the $\geq$1 km detection range; the camera gives vision terminal guidance within a 0.5 kg budget. Onboard radar is not budgeted.
Capture	CO_2 net launcher, 4 m net	Non-destructive, so the target can be recovered intact. 5–15 m is a design target, not a demonstrated range.
Recovery	Tow to nearest GPS safe point, then release under a 2.5 m canopy	Canopy on the net or target at capture was rejected: the load drifts with the wind and the landing point is hard to control.
Camera layout	One forward gimbal camera	A downward camera would be blocked by the netted target; landing safety is handled by GPS safe points.
Remote override	None (single radio link)	Manual RC safety-pilot override was considered and dropped by team decision (10 Oct 2026).

Table 3: Interface summary.

From	**To**	**Interface**	**Note**
Radar	GCS laptop	Ethernet or serial	Update rate sets cue refresh (R30 reference: 64 ms to 3 s)
GCS laptop	Ground radio	USB serial, MAVLink	Targets and commands up, telemetry down
Air radio	Flight controller	UART, MAVLink	1 W class radio
Companion	Flight controller	UART, MAVLink 2	Velocity and yaw-rate setpoints (guided mode)
Camera	Companion	USB or MIPI CSI	Forward gimbal camera
Flight controller	ESCs	DShot	Low-level loops, failsafes, geofence stay on the FC
Flight controller	Net / parachute	PWM or relay	Behind arming interlock

Table 4: Component mass ledger (assumed unless marked). Listed items sum to 14.78 kg; sizing uses 18.00 kg.

Item	**kg**	**Item**	**kg**
Airframe (frame, plates, six arm tubes, fasteners)	4.50	Datalink radio and antenna	0.50
Six motors, U8 Lite KV100 (published, 238 g each)	1.43	Net launcher frame and mechanism	0.80
Six ESCs, Alpha 60A	0.78	Net and tether	0.30
Six propellers, G28x9.2 CF	0.60	CO_2 cartridge	0.20
Landing gear	0.80	Parachute and release	0.40
Battery, 12S 8000 mAh (180 Wh/kg)	1.97	Miscellaneous	0.50
Power board, wiring, connectors	0.80	**Sum of listed items**	**14.78**
Flight controller, GPS, IMU	0.40	Integration / growth allowance (unallocated)	3.22
Companion computer	0.30	**Nominal sizing mass**	**18.00**
Camera and gimbal	0.50	Tow phase (18.0 + 5.0 target + 0.5 net/tether)	23.50

Table 5: Ledger versus bare reference masses (reconciliation pending).

Component set	**Ledger (kg)**	**Reference (kg)**	**Difference (kg)**
Motors (6)	1.43	1.428	0.002
Propellers (6)	0.60	0.354	0.246
ESCs (6)	0.78	0.438	0.342
Cube Orange+ bare module (excludes carrier board, GPS)	0.40	0.053	0.347
Jetson Orin Nano dev-kit assembly	0.30	0.174	0.126
RFD900x-class bare modem (ledger includes antenna)	0.50	0.014	0.486

Table 6: Battery options (12S, 44.4 V nominal, 80% usable).

Pack	Nominal	Usable	Intercept only (207 Wh)	With 60 s tow (253 Wh)
6000 mAh	266 Wh	213 Wh	3% margin (marginal)	Not enough
8000 mAh (selected)	355 Wh	284 Wh	37% margin	12% margin

Table 7: Final dimensions and compliance with the problem statement.

Requirement / parameter	Design value	Status
Sprint ≥25 m/s	3.2 of 7.3 kgf per motor needed at 25 m/s	Meets on paper, not demonstrated
Takeoff weight <25 kg	18.0 kg (23.5 kg with target)	Met
Net capture, parachute recovery ≤5 kg	4 m net, 2.5 m canopy	Design target, not demonstrated
Detection ≥1 km, stated sensor	Ground radar (R30 class)	Sensor stated; range to confirm from data sheet
Vision or radar terminal guidance	Onboard camera and tracker	Design choice
Span / arm / body / height	2.31 m / 0.80 m / about 300 mm / about 450 mm	Computed; body and height assumed
Arm safety factor / tip deflection	10.9 / 6.4 mm	Simplified cantilever check

Table 8: Simulation scenarios.

ID	Scenario	Init. range (m)	Target (m/s)	Key set-up / outcome
1	Attack, slower target (tail chase)	2879	14.87	Candidate outside safe areas, tow required
2	Defense, faster target, lead point	2354	27.0	Candidate inside safe area, release
3	Tow required	2729	12.0	Nearest safe point 225 m away, tow
4	Abort (gate only)	1640	20.0	15.0 s available against 33.8 s needed
5	PN diagnostic (in flight)	1342	12.0	Heading offset $-26.6°$, 20 m/s start
6	Outer ring only (gate only)	4104	20.0	Predicted closest approach 1,800 m
7	Overweight target (6 kg)	2729	12.0	Parachute-request flag, towing disabled
8	Extended pursuit	3551	27.0	Target exits the 1 km zone; energy guard

Table 9: Results for the flown cases (energy values are assumption-based estimates, not validated).

ID	t_{cand} (s)	R_{cand} (m)	V_{close} (m/s)	Sim. (Wh)	Projected (Wh)	Ceiling (Wh)	Outcome
1	88.45	13.35	33.31	120.26	—	253	Tow incomplete at 60 s
2	73.55	14.60	26.97	42.72	79.58	207	Release in safe area
3	82.85	13.97	33.99	85.18	103.98	253	Safe area entered after 13.3 s
5	81.35	14.28	16.09	78.77	—	—	PN diagnostic only
7	82.85	13.97	33.99	76.04	—	207	Parachute request, no tow
8	not registered	13.95	—	78.44	307.84	253	Energy guard stop

Table 10: Thrust margin at 25 m/s under assumed forward-flight de-rating (assumed sensitivity, not measured data; post-capture mass 23.0 kg as in the simulation).

Assumed de-rating	Available thrust (N)	Margin, nominal 18 kg	Margin, post-capture
0% (static)	429.7	2.27	1.82
15%	365.2	1.93	1.55
20%	343.7	1.81	1.46
30% (headline case)	**300.8**	**1.59**	**1.27**

Table 11: Bill of equipment, one drone, no spares (prices in INR including GST where listed; retailer listings checked 10 Oct 2026). S = sourced listing, S† = nearest sourced substitute, E = estimate.

#	Category	Item / specification	Qty	Unit	Total	Basis
1	Propulsion	Motor, T-Motor U8II KV100 (nearest sourced to U8 Lite KV100)	6	33,588	201,528	S†
2	Propulsion	Propeller, T-Motor G28x9.2 CF (pairs)	3	31,004	93,012	S
3	Propulsion	ESC, T-Motor Flame 80A 12S V2.0 (for Alpha 60A)	6	11,195	67,170	S†
4	Power	Battery, 12S 8000 mAh LiPo	1	27,000	27,000	E
5	Power	Power board, XT90, connectors, wire	1	3,500	3,500	E
6	Avionics	Cube Orange+ Standard Set (GPS/compass, ADS-B carrier)	1	34,999	34,999	S
7	Avionics	Companion computer, Jetson Orin Nano class	1	42,000	42,000	E
8	Avionics	Skydroid C12 3-axis gimbal camera	1	53,728	53,728	S
9	Avionics	Long-range radio, RFD900x class (air side)	1	42,000	42,000	E
10	Structure	CF arm tube 35 mm OD × 2 mm, 4.8 m total	4.8 m	1,000	4,800	E
11	Structure	Frame plates, standoffs, fasteners	1	6,000	6,000	E
12	Structure	Landing gear (fixed CF legs)	1	3,500	3,500	E
13	Capture	Net launcher frame and CO_2 firing mechanism	1	8,000	8,000	E
14	Capture	Net and tether, 4 m opening	1	3,000	3,000	E
15	Capture	CO_2 cartridge and holder	1	300	300	E
16	Capture	Parachute, 2.5 m canopy, servo release	1	4,500	4,500	E
17	Misc.	Fasteners, heat-shrink, printed mounts, foam	1	3,000	3,000	E
		Total (one drone, no spares)			**598,037**	

Table 12: Main risks and open items.

Risk / open item	Mitigation or status
3.22 kg unallocated mass; propeller and ESC masses unverified	Assign to real parts or re-run at lower mass; confirm supplier masses before the final CAD freeze
Forward-flight thrust loss unknown	15–30% sensitivity, headline margin 1.59 / 1.27 above 1.0; confirm with eCalc or test
Assumed drag ($C_D = 1.2$, 0.15 m^2)	Replace with CFD or wind-tunnel values
Battery not selected; sag at about 81 A (10.2C)	Choose a pack with datasheet mass and discharge rating; tow margin is only 12%
Energy-limited pursuit (scenario 8: 307.8 Wh against 253 Wh)	Energy guard stops pursuit; place safe points within a 60 s tow of the engagement area
Capture and recovery physics not modelled; no capture-speed envelope	Language limited to geometric candidates; physical testing out of scope here
6 kg target exceeds the 5.5 kg parachute design load	Simulation flag only; a rated canopy, load path and opening-shock check are needed
Single radio link, no RC override	Flight-controller failsafes; fire permission expires on lost heartbeat; lost-link action *[define]*
Thrust falls about 13.6% at 1,500 m altitude	Raise rotor speed; check against motor data
Radio band legality in India	Confirm band and power before choosing the radio (865–867 MHz, 2.4, 5.8 GHz)
Motor-failure recovery and 2.31 m frame acceptability	Not assessed; depends on control authority, not summed thrust